

# Ahmed Ikram | AI & ML Engineering

Dundee, Scotland | +44 7421 996048 | [ahmedikram30@gmail.com](mailto:ahmedikram30@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Online Portfolio](#)

## PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Built a production SageMaker-deployed anomaly detection engine (99.8%  $\pm$  0.1% StratifiedKFold CV accuracy, 97.3% IF precision) and an Agentic RAG assistant cutting inference latency from 27-55s to 11-23s - both validated across 1,402 tests spanning ML, RAG, Kafka, security, and Terraform tiers. Seeking a graduate ML Engineering, AI Engineering, or MLOps role to contribute production-grade model pipelines and LLM system design.

## EDUCATION

### High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

### University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

## PROJECTS

**ATM Log Aggregation, Analysis & Diagnostics Platform** | *FastAPI, XGBoost, Isolation Forest, MLflow, SageMaker, LangChain, ChromaDB, Kafka, Redis, Docker, AWS* | Mar 2026 - June 2026

- Backend and ML engineering lead on a 7-person Agile team for NCR Atleos; independently built the 3-layer detection engine, Agentic RAG assistant, full MLOps infrastructure, AWS SageMaker production inference endpoint, and a 1,402-test suite - all post-deadline on top of the team delivered foundation
- 3-layer anomaly detection engine runs every 30s ML\_Ensemble (XGBoost + Isolation Forest) on 49 hand-engineered features across 7 groups achieving 99.8%  $\pm$  0.1% StratifiedKFold CV accuracy across 8 classes and 97.3% IF precision via grid search and Youden's J threshold calibration (F1=0.7008 at -0.51009); ZScore detects distribution shift via rolling 20-window Z-scores independently cross-checked against a deployed SageMaker endpoint (laad-xgb-champion, ml.t2.medium) for cloud-side validation without making inference dependent on cloud availability; all runs and 7 artifacts versioned in MLflow (RDS backend + S3 artifact store with champion aliases)
- Agentic RAG diagnostic assistant built over ChromaDB + Ollama with cross-encoder reranking (+5–15% retrieval relevance over bi-encoder cosine), self-consistency scoring (3 parallel LLM samples via ThreadPoolExecutor, 3-gram Jaccard similarity), reflexion self-critique (generate  $\rightarrow$  critique  $\rightarrow$  regenerate), and 4-signal confidence fusion (retrieval 30% + consistency 25% + verbalized 25% + grounding 20%) with Platt calibration on ECE trigger; parallelising self-consistency cut end-to-end latency from 26–53s to 11-23s; responses served at <100ms from Redis cache
- 1,402 tests (923 pytest backend at 82% coverage across 10 tiers, 394 vitest frontend, 10 Playwright E2E, 75 Terraform assertions) gating every PR - catching 7 critical defects including a JWT privilege escalation vulnerability and silent data loss under 50 concurrent writes; full AWS estate provisioned via Terraform (10 modules, 118 resources) with 3 GitHub Actions pipelines (OIDC auth, zero static credentials)

**StockLens FinTech Spending & Investment App** | *React Native Expo, TypeScript, Node.js, Python, FastAPI, scikit-learn* | Oct 2025 - Dec 2025

- Implemented an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI alongside the OCR receipt pipeline

**Haggis Species Classification & Predictive Modelling** | *Python, Scikit-learn, XGBoost, Matplotlib, Seaborn, Pandas, NumPy, Jupyter Notebook* | Nov 2025 - Dec 2025

- Engineered and validated 2 novel ratio-based features that emerged as top-3 predictors across all models; prevented data leakage using sklearn Pipelines with ColumnTransformer and RobustScaler throughout
- Trained and compared 7 classifiers including XGBoost, Random Forest, and Logistic Regression achieving ~90% accuracy; tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering to validate natural groupings against ground-truth labels; built a Linear Regression model predicting body mass ( $R^2=0.756$ ), validating assumptions via residual diagnostics

**DevSync Project Tracker with GitHub Integration** | *React, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, CI/CD* | Jan 2025 - Apr 2026

- Deployed production AWS infrastructure: ECS Fargate + ALB + RDS private subnet + S3 + CloudFront + ACM; rolling deploys tagged by commit SHA for immutable rollback; OIDC federation scoped to main branch with no static credentials stored or transmitted anywhere in the pipeline, repository or task definitions

- 1,452 automated tests at 85%/85% line coverage hard-gate every PR - 7 path-aware CI jobs (lint: ruff + ESLint, security: pip-audit + npm audit, Postgres 15 integration, Cypress E2E) must all pass before deployment. Docker layer caching. Dependabot + CodeQL on weekly cadence

**CineMatch AI / ML Movie Recommendation System** | *Python, scikit-learn, Pandas, Flask, NumPy* | *May 2025 - July 2025*

- Implemented three recommendation approaches - collaborative filtering, content-based filtering, and a hybrid model - on the MovieLens dataset; collaborative filtering achieved ~78% hit rate and ~0.22 Precision@10
- Systematically compared all three approaches across hit rate, Precision@10, and cold-start handling - hybrid model outperformed standalone methods by combining user similarity signals with item content features, with each strategy independently swappable via dependency injection for controlled benchmarking

**TECHNICAL SKILLS**

|                |                                                                              |
|----------------|------------------------------------------------------------------------------|
| Languages      | Python · SQL · TypeScript · JavaScript · Bash                                |
| ML & AI        | Scikit-learn · XGBoost · Isolation Forest · Pandas · NumPy                   |
| MLOps          | MLflow · SageMaker · Terraform · Docker · GitHub Actions                     |
| LLM & RAG      | LangChain · ChromaDB · Ollama · ms-marco-MiniML-L-2-v2                       |
| Data & Backend | Apache Kafka · FastAPI · Flask · PostgreSQL · Redis · MySQL · MongoDB        |
| Cloud & Tools  | AWS (ECS Fargate, EC2, S3, RDS, IAM, SageMaker, CloudFront, Secrets Manager) |
| Visualisation  | Matplotlib · Seaborn · Jupyter Notebook · Power BI                           |
| Testing        | Pytest · Vitest · Playwright · dbt                                           |

**CERTIFICATES**

|                                                                            |                 |
|----------------------------------------------------------------------------|-----------------|
| <u>AWS Academy Graduate - Machine Learning Foundations</u>                 | <i>Feb 2025</i> |
| <u>AWS Academy Graduate - Cloud Foundations</u>                            | <i>Mar 2024</i> |
| <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | <i>Oct 2024</i> |

**EXPERIENCE**

|                                                                                                                                                                                      |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Beaully Mini Market</b> , Dundee, Scotland                                                                                                                                        |                                   |
| <i>Retail Assistant</i>                                                                                                                                                              | <i>September 2023 - Present</i>   |
| <ul style="list-style-type: none"> <li>• Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption</li> </ul>                            |                                   |
| <b>Zulus Peri Peri Grill</b> , Dundee, Scotland                                                                                                                                      |                                   |
| <i>Front of House Worker</i>                                                                                                                                                         | <i>June 2021 - September 2023</i> |
| <ul style="list-style-type: none"> <li>• Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure</li> </ul> |                                   |